FORCe window-length pilot

Pipeline v2 (1 s) versus v2.1 (2 s)

Two participants: **spe30 and mle01**, pre/post/retest. All 29 eligible task recordings were included; seven additional recordings had no event-validated trials.

Single change: two independently cleaned 1-second halves versus one full 2-second epoch. Both used two decomposition levels, 500 Hz EEG, 1-20 Hz filtering before FORCe, the same baseline and rejection rules. These pilots preceded the later filter-order and coding updates.

Plot key: dashed = v2 / 1 s; solid = v2.1 / 2 s. Blue = non-displaced; red = displaced. All following plots use trials accepted by both versions (2,610 total), so trial selection is matched.

Measure	v2: 1 s	v2.1: 2 s
Median onset step (microvolts)	2.794	0.571
Median nearby step (microvolts)	0.586	0.582
Median within-trial onset / nearby ratio	4.560	0.984
Total accepted trials (each version)	2,613	2,611

Onset step: for one trial, the maximum absolute change across channels between the samples at -2 ms and 0 ms. **Median onset step:** the median of those per-trial values across all 2,610 matched trials.

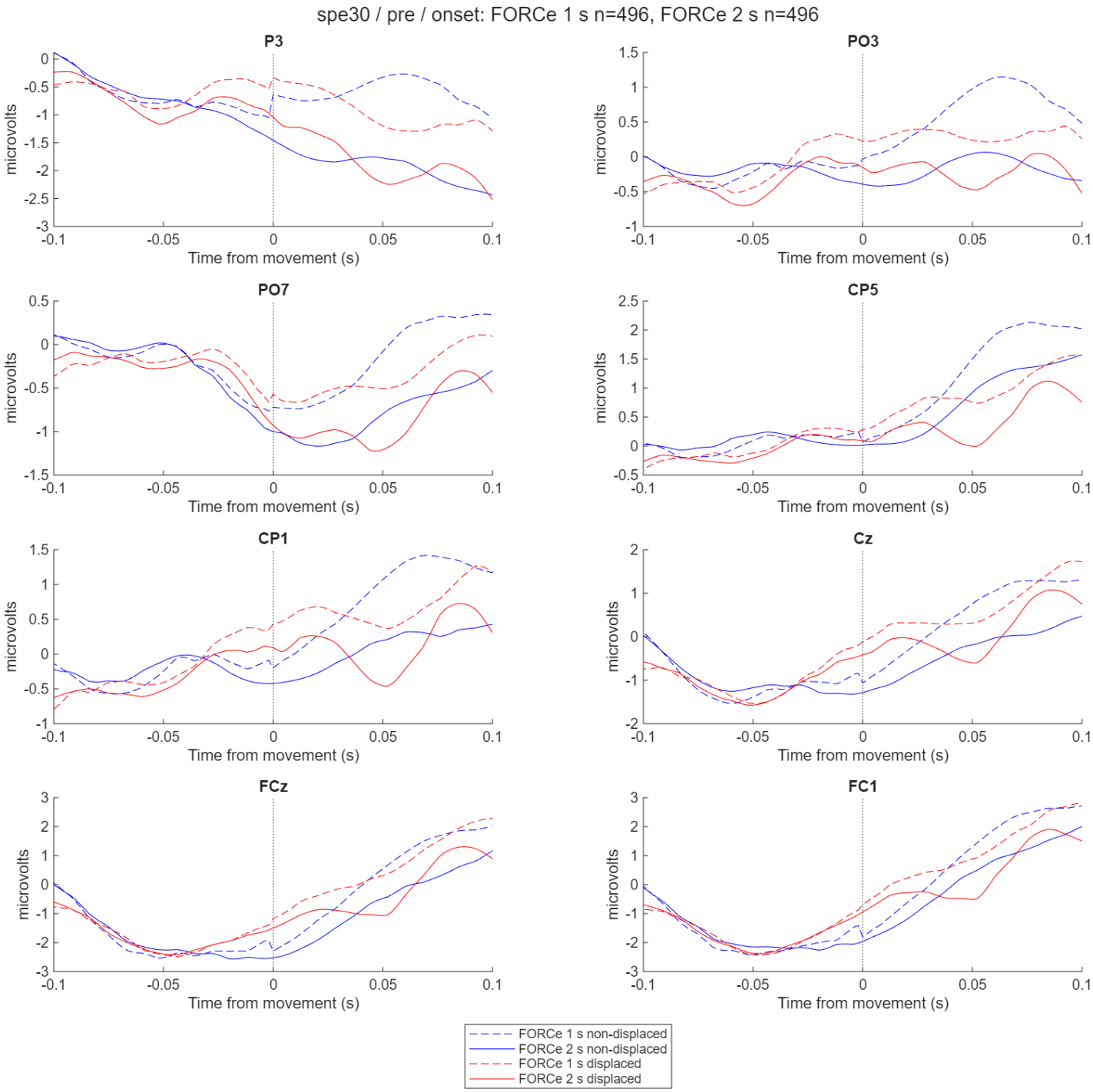
Nearby step: for one trial, the median of maximum-channel adjacent-sample changes within +/-50 ms, excluding the onset pair. The table reports its median across trials, and the median of the per-trial onset / nearby ratios.

Interpretation: the median onset step was 79.6% lower, while nearby sample changes were similar. This supports reduced boundary discontinuity; it does not establish equivalent or superior artifact removal.

Pages 2-13 show each participant/session as a movement-onset detail followed by the full-epoch view. The PDF presents existing pilot results; preparing it did not change processing or thresholds.

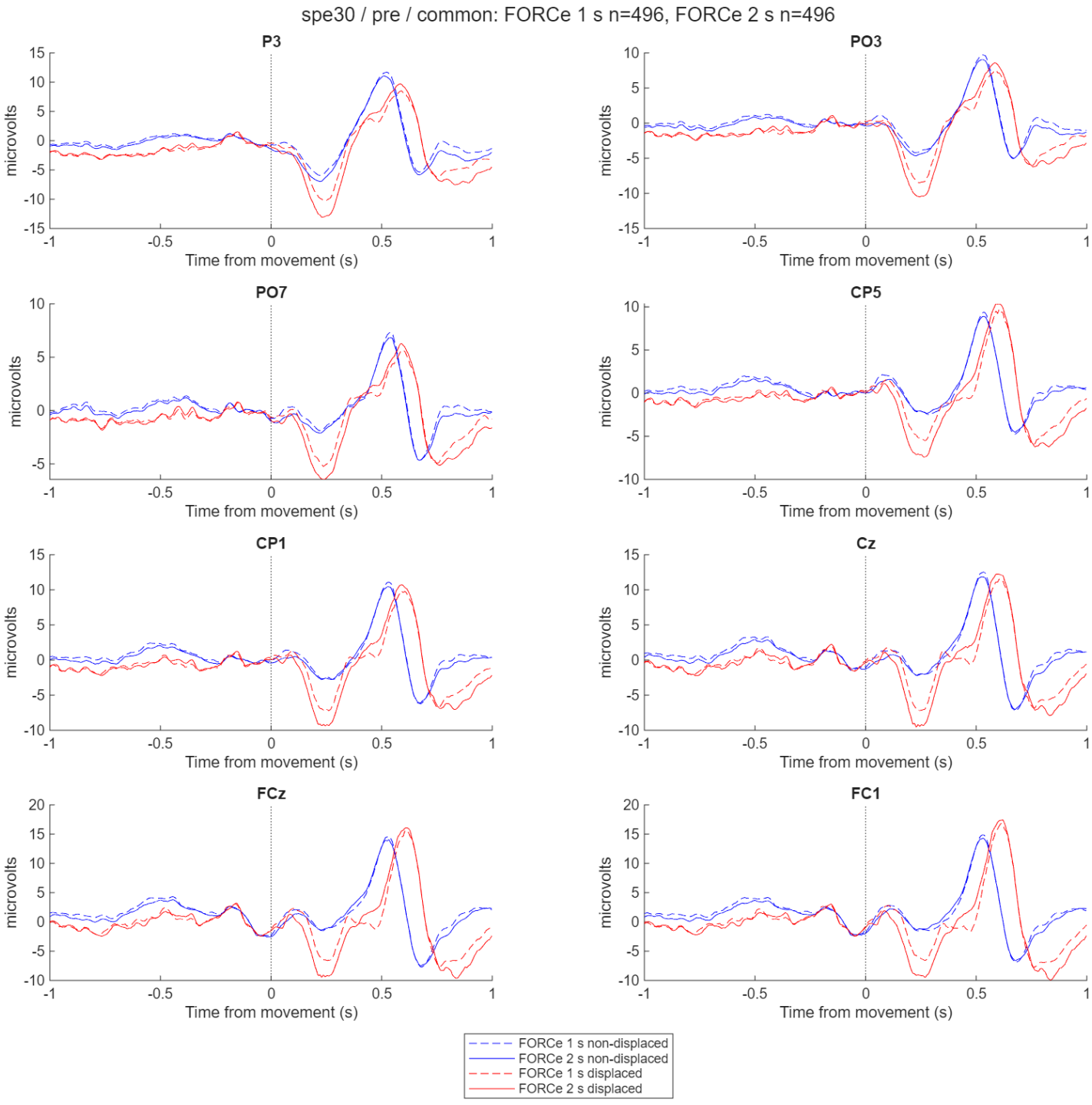
spe30 / pre

Movement-onset detail: -0.1 to +0.1 s



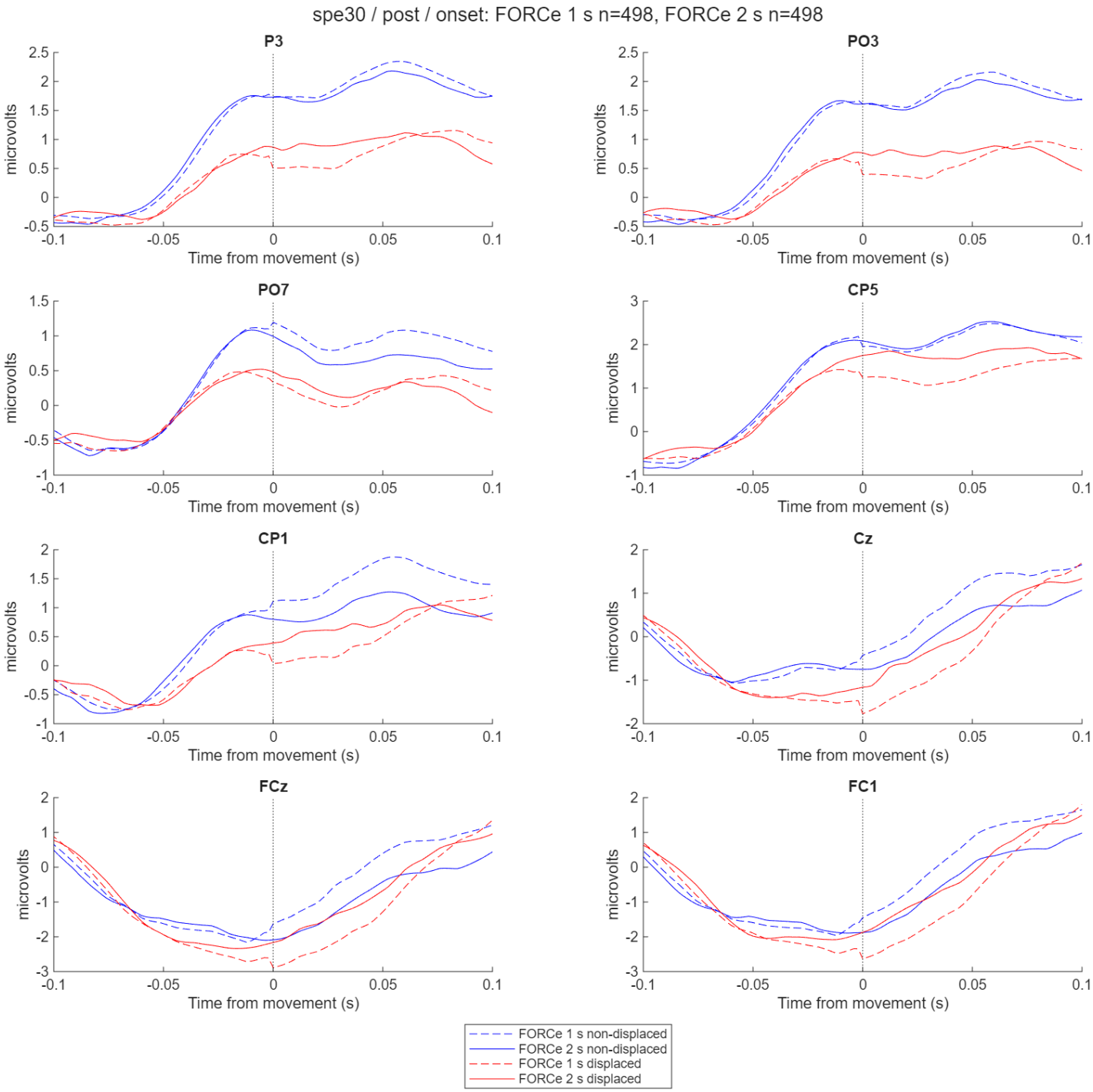
spe30 / pre

Full epoch: -1 to +1 s



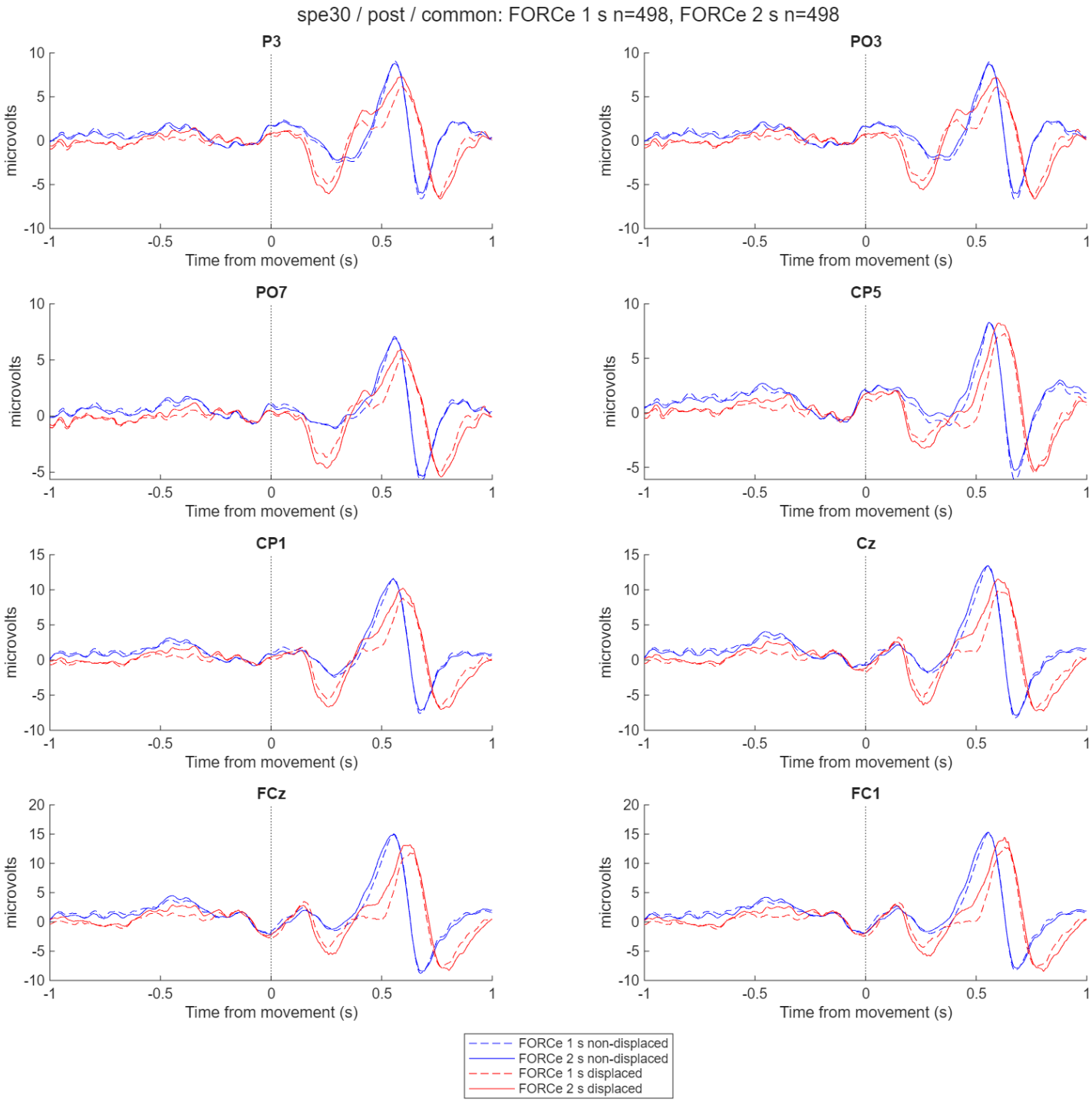
spe30 / post

Movement-onset detail: -0.1 to +0.1 s



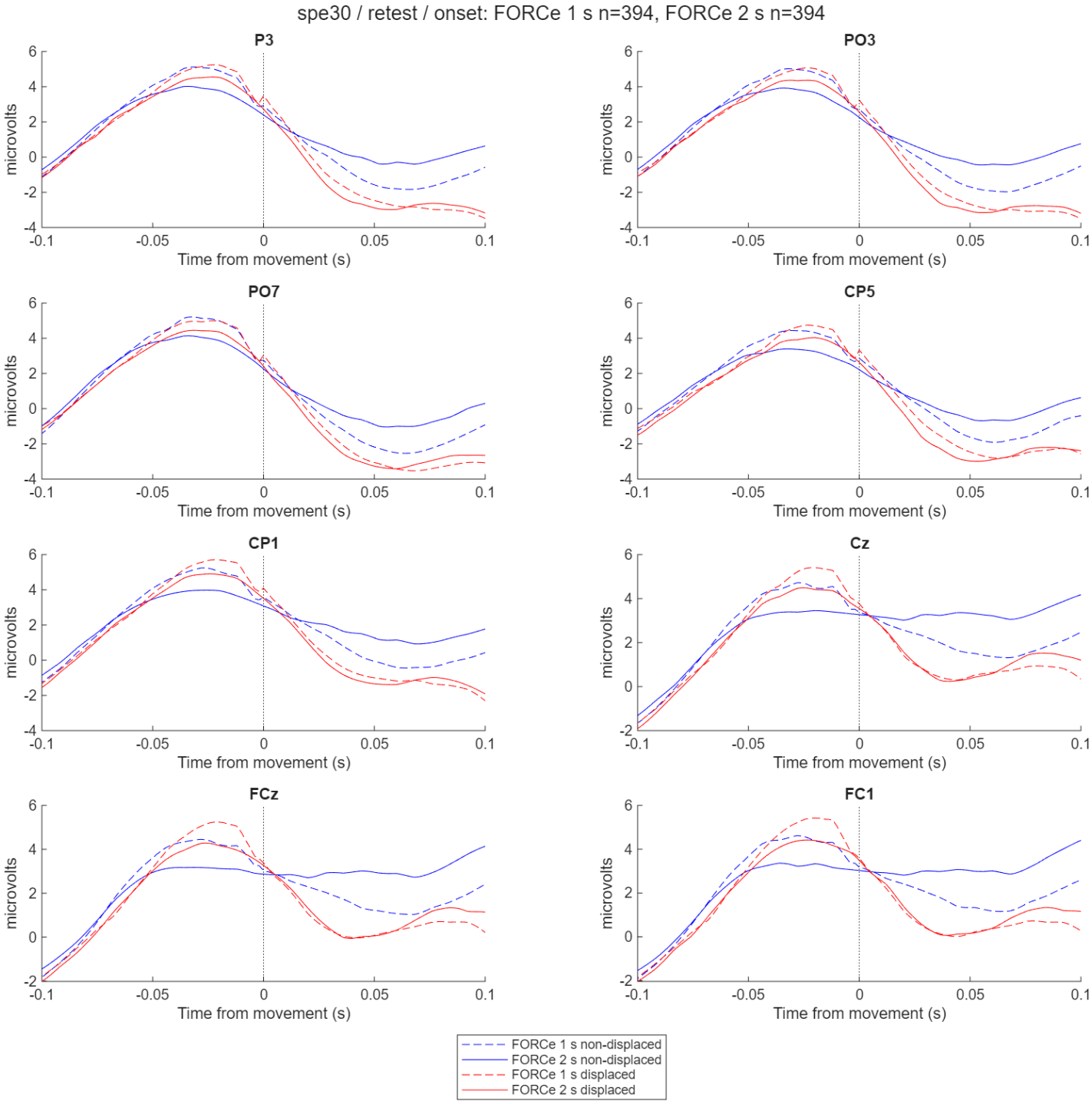
spe30 / post

Full epoch: -1 to +1 s



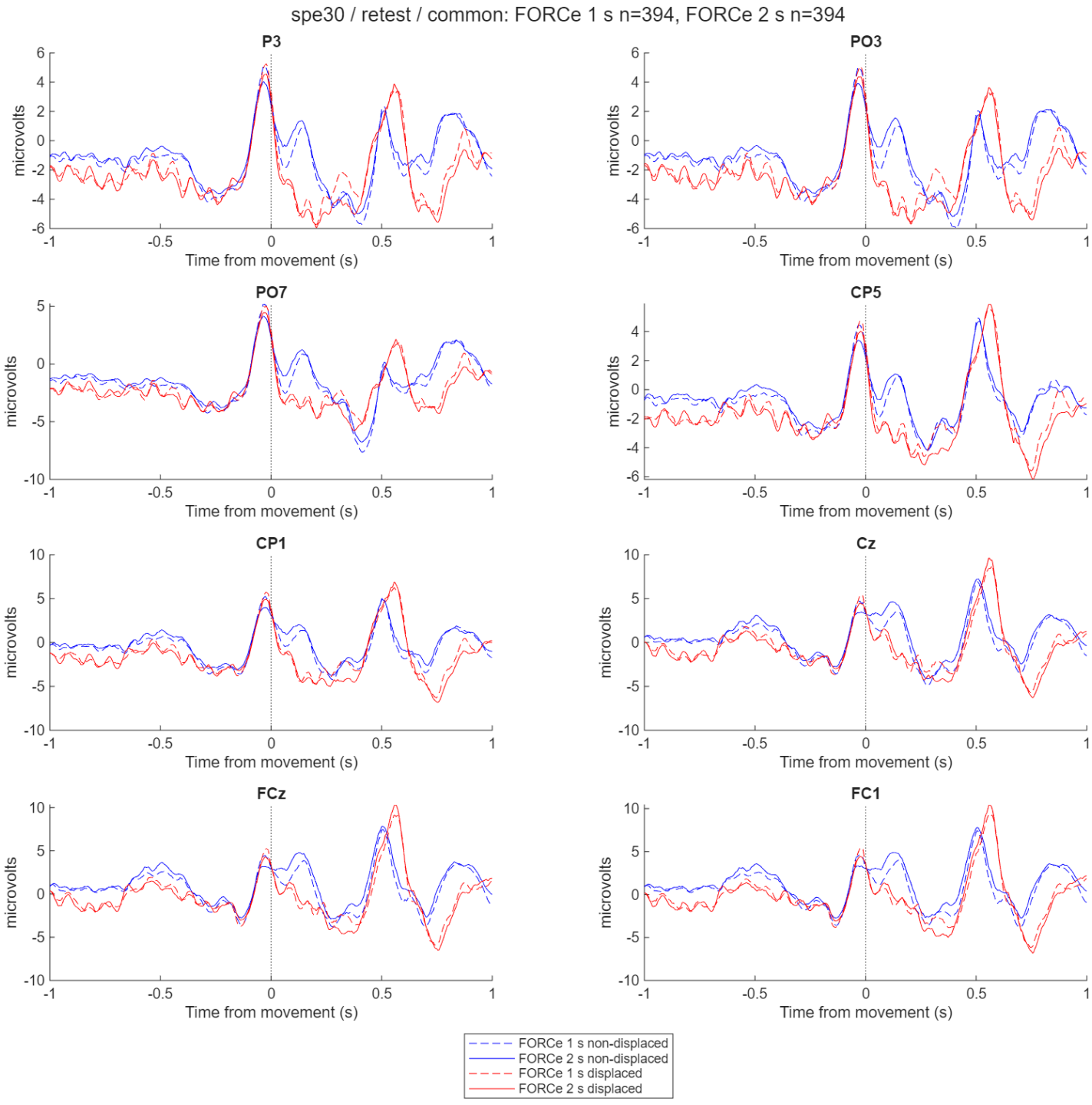
spe30 / retest

Movement-onset detail: -0.1 to +0.1 s



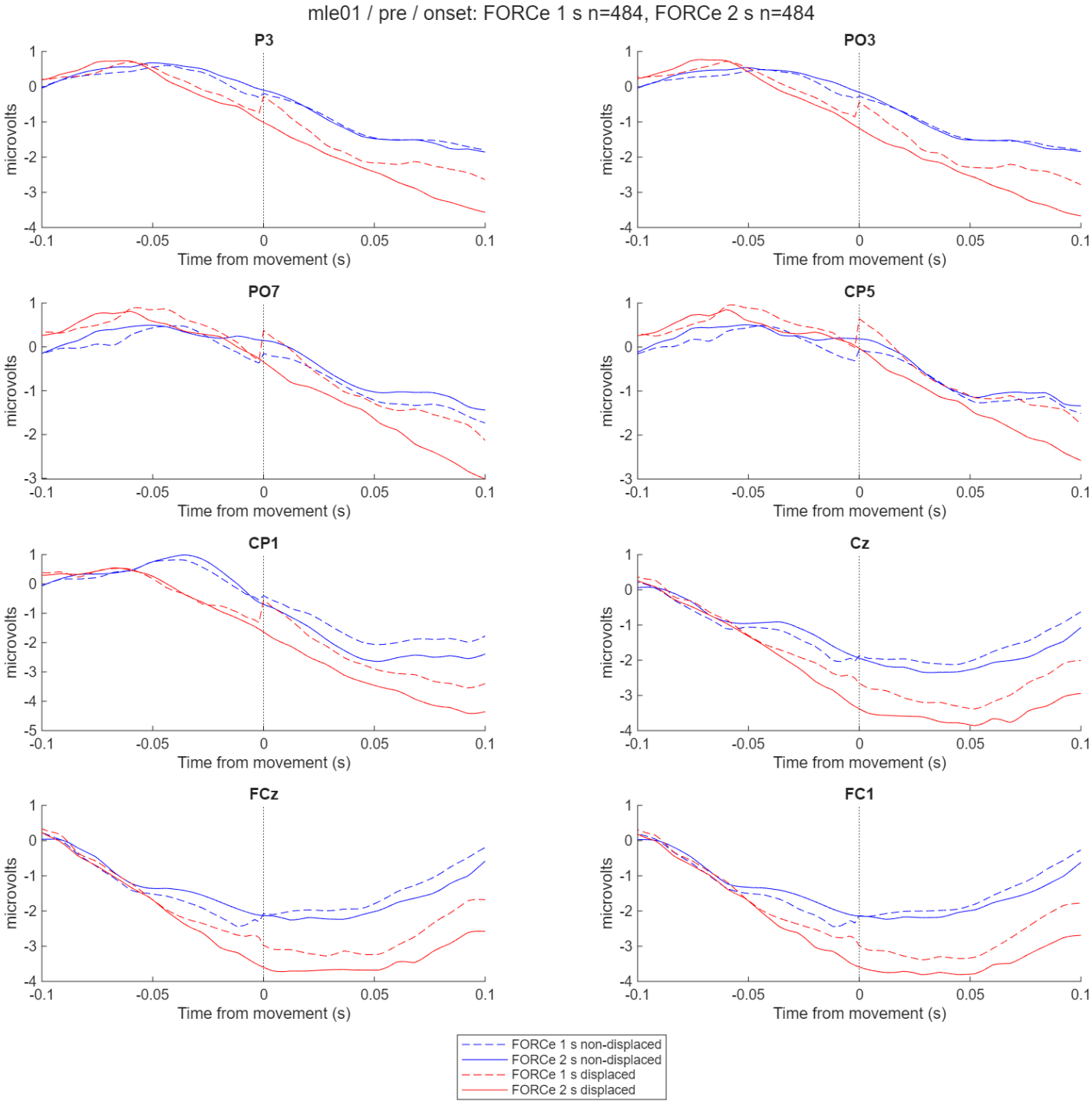
spe30 / retest

Full epoch: -1 to +1 s



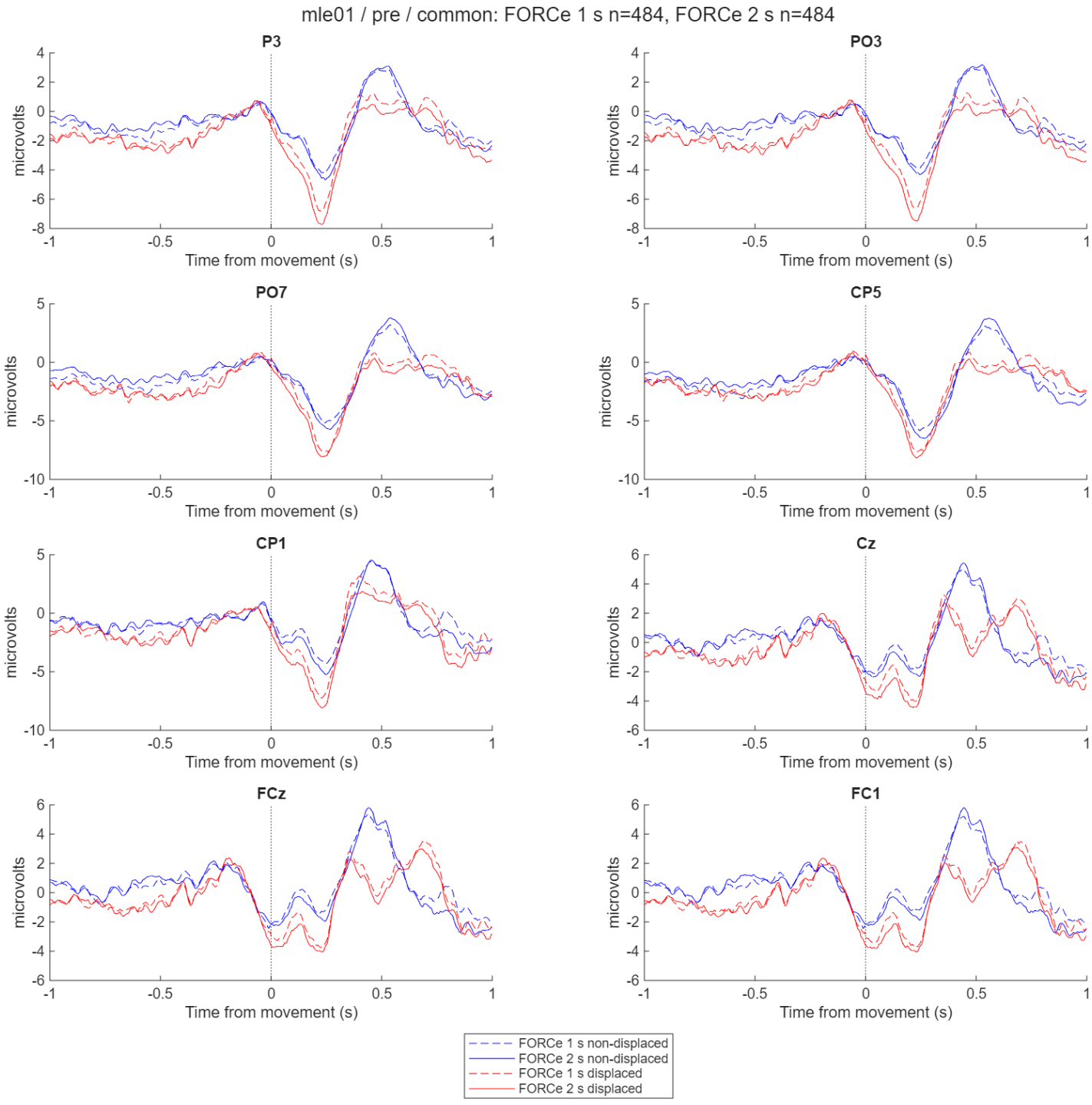
mle01 / pre

Movement-onset detail: -0.1 to +0.1 s



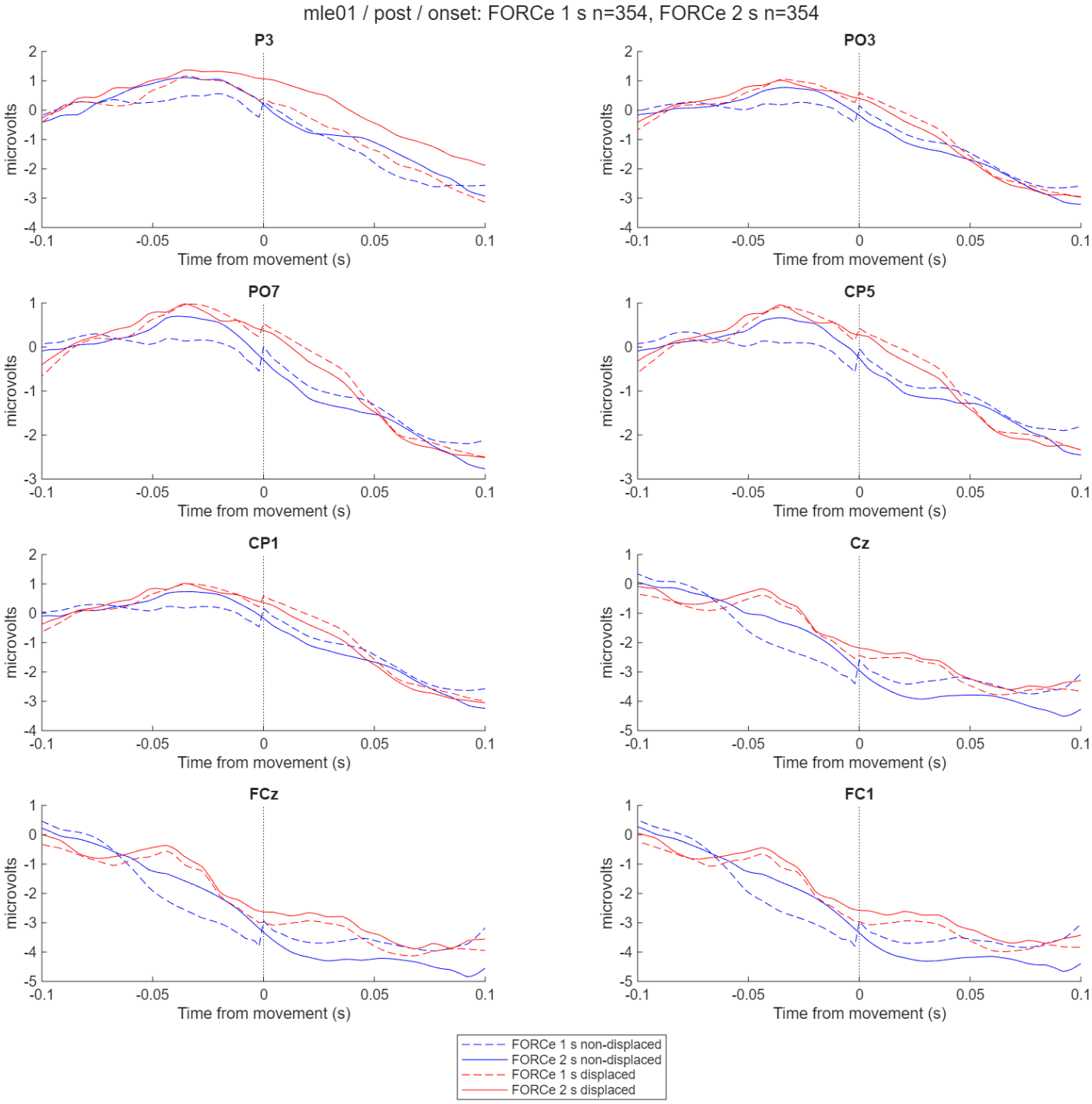
mle01 / pre

Full epoch: -1 to +1 s



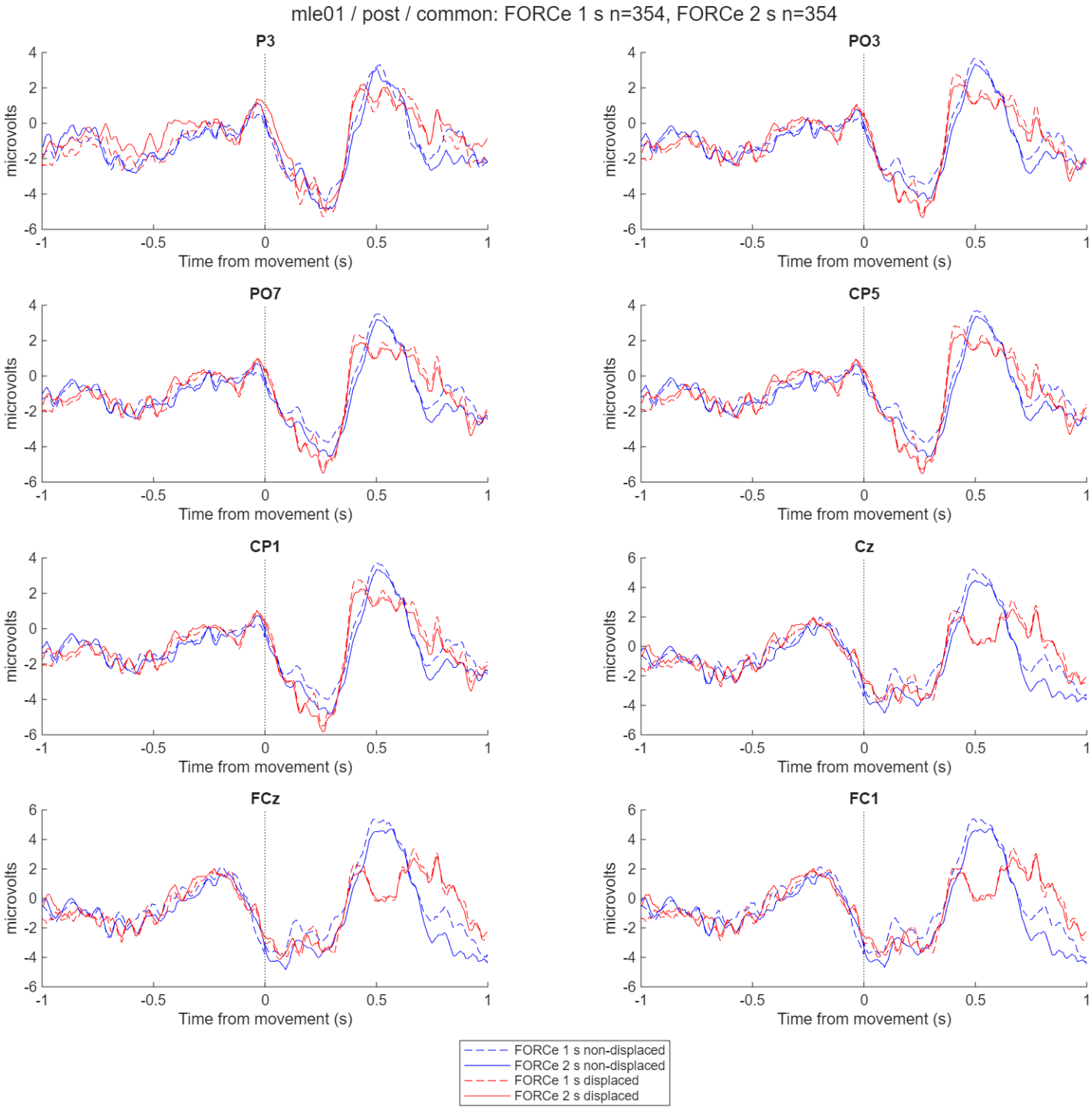
mle01 / post

Movement-onset detail: -0.1 to +0.1 s



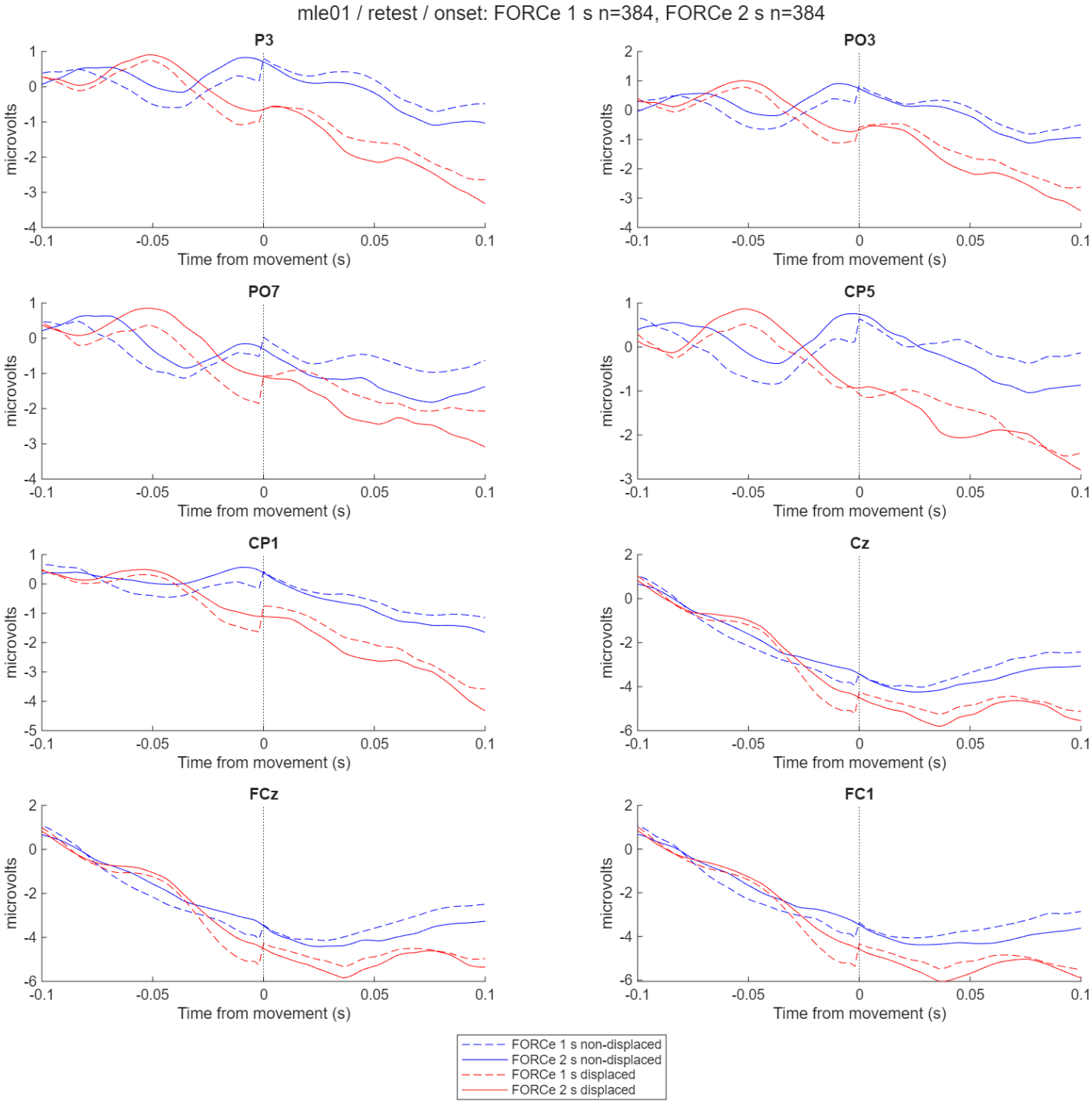
mle01 / post

Full epoch: -1 to +1 s



mle01 / retest

Movement-onset detail: -0.1 to +0.1 s



mle01 / retest

Full epoch: -1 to +1 s

